Xiaowei Ren

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Professional Experience

Jun. 2021 – Present	Senior Architect	NVIDIA, Canada and USA and China
Nov. 2020 – Jun. 2021	Postdoc Research Fellow	University of British Columbia, Canada
Sept. 2015 – Oct. 2020	Research Assistant	University of British Columbia, Canada
Sept. 2019 – Nov. 2019	Research Intern	Max Planck Institute for Software Systems, Germany
Aug. 2018 – Nov. 2018	Research Intern	NVIDIA, USA
May. 2017 – Aug. 2017	Research Intern	NVIDIA, USA
Sept. 2012 – Jun. 2015	Research Assistant	Xi'an Jiaotong University, China
Jul. 2011 – Sept. 2011	Undergraduate Intern	ICT, Chinese Academy of Science, China

Education

Sept. 2015 – Oct. 2020 PhD in Computer Engineering University of British Columbia, Canada

Dissertation: Efficient Synchronization Mechanisms for Scalable GPU Architectures

Sept. 2012 – Jun. 2015 M.Sc. in Computer Engineering Xi'an Jiaotong University, China

Thesis: Parallel Acceleration Algorithms and FPGA Implementation for KLMS and KAP

Sept. 2008 – Jun. 2012 B.Sc. in Electronic Engineering Xi'an Jiaotong University, China

Publications

- Michalis Kokologiannakis, **Xiaowei Ren**, and Viktor Vafeiadis. "Dynamic Partial Order Reductions for Spinloops", *Annual International Conference of Formal Methods in Computer-Aided Design (FMCAD)*, Yale University, Connecticut, USA, October 2021.

- **Xiaowei Ren**, and Mieszko Lis. "CHOPIN: Scalable Graphics Rendering in Multi-GPU Systems via Parallel Image Composition", *27th International Symposium on High Performance Computer Architecture (HPCA)*, Seoul, South Korea, February 2021. (acceptance rate: 63/258 = 24.4%)

- Dingqing Yang, Amin Ghasemazar*, **Xiaowei Ren***, Maximilian Golub, Guy Lemieux, and Mieszko Lis. "Procrustes: a Dataflow and Accelerator for Sparse Deep Neural Network Training", *53rd International Symposium on Microarchitecture (MICRO)*, Athens, Greece, October 2020. (acceptance rate: 82/424 = 19.3%, *equal contribution)

- **Xiaowei Ren**, Daniel Lustig, Evgeny Bolotin, Aamer Jaleel, Oreste Villa, and David Nellans. "HMG: Extending Cache Coherence Protocols Across Modern Hierarchical Multi-GPU Systems", *26th International Symposium on High Performance Computer Architecture (HPCA)*, San Diego, USA, February 2020. (acceptance rate: 48/248 = 19.4%)

- **Xiaowei Ren**, and Mieszko Lis. "High-Performance GPU Transactional Memory via Eager Conflict Detection", *24th International Symposium on High Performance Computer Architecture (HPCA)*, Vienna, Austria, February 2018. (acceptance rate: 54/260 = 20.8%)

- **Xiaowei Ren**, and Mieszko Lis. "Efficient Sequential Consistency in GPUs via Relativistic Cache Coherence", *23rd International Symposium on High Performance Computer Architecture (HPCA)*, Austin, USA, February 2017. (acceptance rate: 50/224 = 22.3%)

- Pengju Ren, **Xiaowei Ren**, Sudhanshu Sane, Michel A. Kinsy, and Nanning Zheng. "A Deadlock-Free and Connectivity-Guaranteed Methodology for Achieving Fault-tolerance in On-Chip Networks", *IEEE Transactions on Computers (TC)*, 2016.

- **Xiaowei Ren**, Qihang Yu, Badong Chen, Nanning Zheng, and Pengju Ren. "A Reconfigurable Parallel Accelerator for the Kernel Affine Projection Algorithm", *IEEE International Conference on Digital Signal Processing (DSP)*, Singapore, July 2015.

- **Xiaowei Ren**, Qihang Yu, Badong Chen, Nanning Zheng, and Pengju Ren. "A 128-way FPGA Platform for the Acceleration of KLMS Algorithm", *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan, January 2015. (University LSI Design Contest)

- **Xiaowei Ren**, Qihang Yu, Badong Chen, Nanning Zheng, and Pengju Ren. "A Real-time Permutation Entropy Computation for EEG Signals", *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan, January 2015. (University LSI Design Contest)

- **Xiaowei Ren**, Pengju Ren, Badong Chen, Jose C. Principe, and Nanning Zheng. "A Reconfigurable Parallel Acceleration Platform for Evaluation of Permutation Entropy", *36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Chicago, USA, August 2014.

- **Xiaowei Ren**, Pengju Ren, Badong Chen, Tai Min, and Nanning Zheng. "Hardware implementation of KLMS Algorithm using FPGA", *International Joint Conference on Neural Networks (IJCNN)*, Beijing, China, July 2014.

- Pengju Ren, Qingxin Meng, **Xiaowei Ren**, and Nanning Zheng. "Fault-tolerant Routing for On-chip Network without Using Virtual Channel", *ACM/EDAC/IEEE Design Automation Conference (DAC)*, San Francisco, USA, June 2014. (acceptance rate: 3150/10963 = 29%)

Talks & Presentations

- Oral, "CHOPIN: Scalable Graphics Rendering in Multi-GPU Systems via Parallel Image Composition", *HPCA*, Global Virtual Event, February 2021.

- Oral, "HMG: Extending Cache Coherence Protocols Across Modern Hierarchical Multi-GPU Systems", *HPCA*, San Diego, USA, February 2020.

- Oral, "High-Performance GPU Transactional Memory via Eager Conflict Detection", *HPCA*, Vienna, Austria, February 2018.

- Oral, "Efficient Sequential Consistency in GPUs via Relativistic Cache Coherence", *HPCA*, Austin, USA, February 2017.

- Oral and Poster, "A 128-way FPGA Platform for the Acceleration of KLMS Algorithm", *ASP-DAC*, Tokyo, Japan, January 2015.
- Oral and Poster, "A Real-time Permutation Entropy Computation for EEG Signals", *ASP-DAC*, Tokyo, Japan, January 2015.
- Poster, "A Reconfigurable Parallel Acceleration Platform for Evaluation of Permutation Entropy", *EMBC*, Chicago, USA, August 2014.
- Poster, "Hardware implementation of KLMS Algorithm using FPGA", *IJCNN*, Beijing, China, July 2014.

Professional Service

- External Reviewer Committee, MICRO, 2021
- External Reviewer Committee, ISCA, 2021, 2022
- Artifact Evaluation Committee, ASPLOS, 2021
- Reviewer, ACM TECS, 2020
- Reviewer, ACM TACO, 2020, 2021, 2022
- Reviewer, IEEE TPDS, 2021
- Reviewer, IEEE TC, 2020, 2022
- Reviewer, IEEE TCAD, 2020, 2021
- Reviewer, IEEE CAL, 2020, 2021
- Reviewer, JPDC, 2020
- Subreviewer, HPCA, 2021
- Shadow Program Committee, EuroSys, 2021

Awards

2016 – 2020	UBC Graduate Support Initiative (GSI) Awards
2012 – 2015	National Master Scholarship (honors top 5% students)
2013 – 2014	Suzhou Industrial Park Scholarship
2010 – 2011	CASC Secondary Class Scholarship
2009 – 2010	National Encouragement Scholarship (honors top 3% students)
2008 – 2009	Siyuan Scholarship

Teaching Experience

Jan. 2017 – Apr. 2017	Teaching Assistant, University of British Columbia, Canada EECE527: Advanced Computer Architecture (Instructor: Mieszko Lis)
Sept. 2016 – Dec. 2016	Teaching Assistant, University of British Columbia, Canada CPEN411: Computer Architecture (Instructor: Mieszko Lis)
Sept. 2015 – Dec. 2015	Teaching Assistant, University of British Columbia, Canada CPEN211: Introduction to Microcomputers (Instructor: Tor Aamodt)